

On the Structural Asymmetry Between General Relativity and Quantum Mechanics: Why Gravity Cannot Be Unified as a Conventional Field

Paul Jarvis¹

¹Independent Researcher, United Kingdom

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Abstract

General Relativity (GR) and Quantum Mechanics (QM) remain mathematically incompatible despite a century of effort. This paper argues that the failure of unification arises from a fundamental structural asymmetry: gravity is not a force within spacetime but the geometric framework that *defines* spacetime. All particles and fields exist within this gravitational structure, making gravity universal in a way no quantum interaction is. Because quantum field theory requires a fixed background while GR denies the existence of such a background, unification cannot proceed by treating gravity as another quantum field. A successful theory must instead emerge from a deeper, pre-geometric quantum substrate from which both spacetime and quantum fields arise. This conceptual insight is formalized here as the *Gravitational Baseline Principle*.

1 Introduction

Einstein's General Relativity (GR) revolutionized physics by identifying gravity with the curvature of spacetime itself [1]. Quantum Mechanics (QM), developed in parallel, describes matter and radiation through probabilistic amplitudes evolving on a fixed spacetime background [2]. Despite their empirical success, the two theories remain conceptually incompatible.

This paper develops a conceptual argument: GR and QM cannot be unified as long as gravity is treated as a field analogous to the electromagnetic, weak, or strong interactions.

Gravity is not a field *within* spacetime; it is the structure that *defines* spacetime. This asymmetry prevents unification at the level of fields and suggests that a deeper, pre-geometric quantum substrate must underlie both theories.

2 Gravity as Universal Geometry

Einstein's field equations,

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}, \tag{1}$$

express the idea that matter and energy determine the curvature of spacetime. Every particle, regardless of charge or composition, follows geodesics determined by this curvature. Gravity couples to all forms of energy and momentum.

By contrast, the quantum interactions are selective:

- Electromagnetism acts only on electrically charged particles.
- The weak interaction acts only on left-handed fermions.
- The strong interaction acts only on quarks and gluons.

Gravity alone is universal. It is not one interaction among many; it is the geometric background in which all interactions occur.

3 The Asymmetry Problem

Quantum field theory (QFT) assumes a fixed spacetime background. GR denies the existence of such a background. This leads to a structural contradiction:

Quantum Mechanics	General Relativity
Requires fixed spacetime	Spacetime is dynamic
Fields evolve <i>in</i> spacetime	Geometry <i>is</i> the field
Linear operators	Nonlinear curvature
Local interactions	Background-dependent

This mismatch is conceptual, not technical. One theory requires a stage; the other *is* the stage.

4 The Gravitational Baseline Principle

Principle (Gravitational Baseline Principle). *Gravity is not a force within spacetime but the geometric structure that defines spacetime. Because all physical systems exist and evolve within this structure, gravity acts as the universal baseline against which all other interactions are measured. Any theory attempting to unify gravity with the quantum forces must therefore operate at a deeper, pre-geometric level where spacetime itself is not fundamental.*

This principle formalizes the structural asymmetry between gravity and the quantum interactions. It asserts that unification cannot succeed at the level of fields because gravity is not a peer to the gauge forces; it is the metric foundation that makes the gauge forces definable.

5 Gravity as the Baseline of Measurement

Gravity defines the fundamental scales of nature:

$$\ell_{\text{P}} = \sqrt{\frac{\hbar G}{c^3}}, \quad (2)$$

$$t_{\text{P}} = \sqrt{\frac{\hbar G}{c^5}}, \quad (3)$$

$$m_{\text{P}} = \sqrt{\frac{\hbar c}{G}}. \quad (4)$$

These Planck units set the baseline against which all other physical quantities are measured. Gravity is therefore not merely another field to unify; it is the *metric foundation* of the entire measurement system.

6 Conceptual Diagram: Gravity as Background

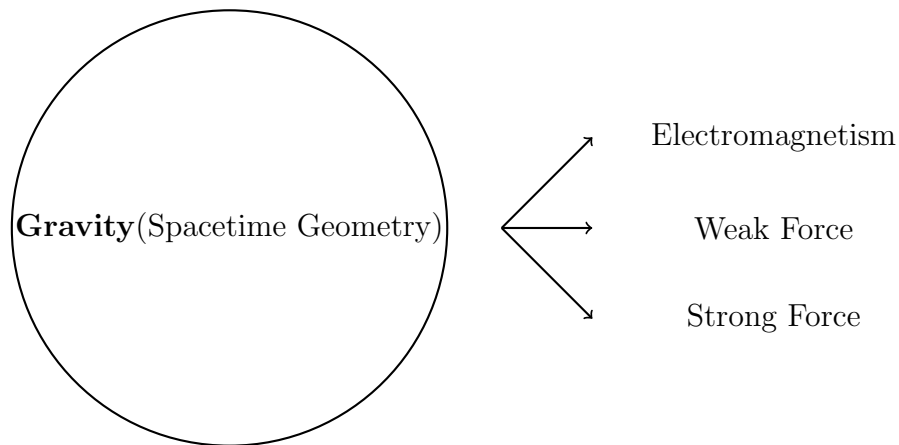


Figure 1: Gravity as the geometric background in which all other interactions occur.

7 Background Independence

GR is background independent: the geometry of spacetime is not fixed but determined dynamically. QFT, by contrast, requires a fixed background metric. This incompatibility is central to the failure of quantization attempts [3, 4].

8 Modern Conceptual Insight: Geometry from Quantum Structure

Recent developments suggest that spacetime geometry may emerge from deeper quantum structure. Entanglement patterns have been shown to encode geometric connectivity in holographic models [5, 6]. This motivates the idea that gravity is emergent rather than fundamental.

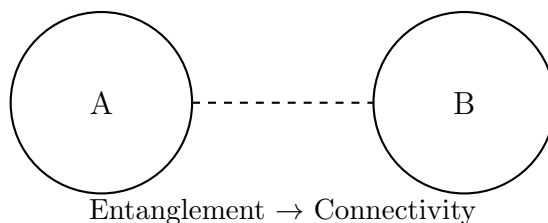


Figure 2: A conceptual diagram illustrating the modern idea that entanglement structure may define geometric connectivity.

9 Implications for Unification

If gravity is the structure of spacetime, then unification cannot proceed by merging gravity with the quantum forces. Instead, unification must arise from a deeper level where:

- spacetime is not fundamental,
- gravity emerges from quantum structure,
- geometry arises from entanglement or other pre-geometric degrees of freedom.

This perspective aligns with emergent spacetime models, holography, and relational approaches to quantum gravity [7].

10 Testable Expectations and Theoretical Constraints

The Gravitational Baseline Principle is a structural claim about the nature of gravity and spacetime. While it does not yet yield sharp numerical predictions, it imposes clear constraints on the form of any viable theory of quantum gravity and leads to broad classes of testable expectations. These expectations arise from the idea that gravity is the geometric baseline rather than a field within spacetime, and that spacetime itself must emerge from deeper, pre-geometric degrees of freedom.

10.1 Absence of a Fundamental Graviton Field

If gravity is not a field within spacetime, then the graviton cannot be a fundamental particle in the same sense as photons or gluons. Any successful theory must therefore avoid treating the metric as a quantizable field on a background. This implies that perturbative quantum gravity and background-dependent graviton models cannot be fundamental descriptions. A decisive detection of a background-propagating graviton would challenge the principle.

10.2 Spacetime as an Emergent Medium

The principle implies that spacetime is not fundamental but emergent. This suggests that at sufficiently small scales, spacetime may exhibit discreteness, nonlocality, or effective cutoffs. Observable consequences may include modified dispersion relations at ultra-high energies, deviations from exact Lorentz invariance, or limits on information density. While current experiments have not reached the Planck scale, astrophysical observations and precision tests of Lorentz symmetry continue to probe these possibilities.

10.3 Gravity as a Thermodynamic Equation of State

If gravity is the baseline structure of spacetime, then Einstein's equations may arise as thermodynamic relations describing coarse-grained microstructure. This perspective is supported by the thermodynamic behavior of black holes and the ubiquity of area laws for entropy. The principle therefore predicts that gravitational dynamics should exhibit robustness characteristic of thermodynamic equations of state, and that small corrections to Einstein's equations may appear as higher-order thermodynamic terms.

10.4 Information and Entropy Bounds as Structural Features

Emergent spacetime models often imply fundamental limits on information density, such as holographic entropy bounds. Under the Gravitational Baseline Principle, such bounds are not incidental but structural: they reflect the finite information capacity of the underlying pre-geometric substrate. Observational confirmation of holographic noise, entropy bounds in cosmology, or universal area laws would be consistent with this expectation.

10.5 Constraints on Unified Theories

The principle imposes a strong constraint: no theory that treats gravity as a gauge field on a fixed background can be fundamentally correct. A viable unification must instead derive both spacetime geometry and quantum fields from a deeper substrate. This excludes broad classes of background-dependent approaches while supporting emergent, relational, and information-theoretic frameworks.

In summary, while the Gravitational Baseline Principle does not yet yield precise experimental predictions, it provides a clear set of theoretical constraints and testable expectations. These guide the search for a deeper, pre-geometric theory from which both spacetime and quantum fields emerge.

11 Conclusion

GR and QM resist unification because they describe physics at different conceptual levels. Gravity is not a force within spacetime; it is the structure that defines spacetime. All particles and interactions occur within this gravitational framework, making gravity the universal baseline rather than a peer to the quantum forces. A successful unification must therefore arise from a deeper, pre-geometric quantum substrate from which both spacetime and quantum fields emerge.

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